

ITAMAR ROCHA FILHO

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Languages Portuguese, English, French, Spanish.

Github : ItamarRocha **Linkedin** : in/itamarrocha **Personal Site** : <https://itamarrocha.github.io>

EDUCATION

Masters of Engineering in Computational Science and Engineering - Harvard University 09/2024 - 05/2026

· TF for AC215 (Machine Learning Operations)

· Research on RLVR advised by Sham Kakade

B.S in Computer Engineering - Universidade Federal da Paraíba (BRAZIL) - 9.59/10 (1st) 07/2018 - 05/2024

RESEARCH

Evolutionary-Alignment In preparation for ICML 2026

a

Metrics for RL performance Ongoing

Work in-progress

Parameter-Efficient Reinforcement Learning Using Prefix Optimization Under Review - ICLR 2026

a

Evaluating Misalignment Under Pressure Mini Experiment for AI Safety Class

a

An Approach for Automatic Description of Characters for Blind People WebMedia 2021

Iris-CV: Classifying Iris Flowers Is Not as Easy as You Thought BRACIS 2021

EXPERIENCE

Researcher - Kempner Institute 05/2025 - current

Conducting research on Reinforcement Learning With Verifiable Rewards on LLMs. Produced one research paper on creating a more computationally and cost-efficient RL method based on prefix optimization. Also working on analyzing post-training metrics and other algorithmic approaches. I've built more than 40 base models and run hundreds of post-training and SFT pipelines in models with parameters ranging from 150M to 8 B.

SWE/MLE - about.webdraw.com (Joined before the seed round) 11/2023 - 06/2024

I developed the company's first AI product: a LLM-based Sales Assistant. Also worked on infrastructure, created a tiered cache system, created the internal load test, sponsored competitions, and hired new individuals. I also created an AI gateway that integrated 20+ different generative image, audio, text, and video providers in a uniform pattern and made a hands-free mode leveraging voice models.

Software Engineer Intern / Research - Google [Mountain View - CA] 06/2023 - 09/2023

I interned at the YouTube Strategic Video Compression team, working on an open-ended research project. The project involved taking different approaches to improve the ultra-popular video compression algorithm. I cut the cost of the component by 50% while also decreasing the model's error by 7% and simplifying the code base to one model instead of six.

Software Engineer Intern - Meta [London - UK] 01/2023 - 04/2023

I worked on the App Ads Platform team. I was responsible for creating a new feature for Instagram Ads. The project consisted of creating an end-to-end demo (Back end + Front end: web and mobile), testing, Analyzing metrics, Resolving bugs, and collaborating with other teams. I completed the project in advance and started working on side projects to help the team.

Software Engineer Machine Learning Intern - Meta [London - UK] 01/2022 - 04/2022

I worked on the AI for AR team. The team is focused on cutting-edge research in computer vision and audio. My internship involved adapting a state-of-the-art model to use Meta's production environment, cleaning, refactoring the code, and deploying it for further testing and production use. I also built a demo for internal use.

Software Engineering Intern - Google [Belo Horizonte - BR] 08/2021 - 11/2021

I worked on the Google Search - contextualization team. My work involved exploring and analyzing a new data source creating an end-to-end demo (Data Generation + Analysis + Back-end + Front-end). After completing my demo, I increased the original feature's coverage by 68.38% while maintaining a high data quality.

TOOLS AND INTERESTS

Tools C, C++, Python, PyTorch, vLLM, OpenRLHF, verl, Ray, Typescript.

Interests Post-training, Algorithms, Optimization, Distributed Training and Infra, Software Engineering, MLOps.

EXTRACURRICULAR ACTIVITIES

Ex-President, Co-Founder and Advisor TAIL - UFPB

08/2020 - 12/2025

Technology and Artificial Intelligence League is a non-profit organization aimed at teaching, producing, contributing to research, and helping members get started on internships, graduation programs, and full-time jobs. It currently has 65 active members and has introduced hundreds of people from multidisciplinary areas to programming and AI.

Founder and Advisor - Momento

08/2024 - Present

Momento is an NGO that focuses on mapping alumni from public universities in Brazil and generating value through mentorship, events, and competitions. We are currently running the second mentorship cohort, with more than 200 student applications from all over Brazil, and mentors from industry and academia, including representatives from MIT, Harvard, DeepMind, Google, Cohere, and many others.

HONORS AND AWARDS

NXTP Fellow

11/2025

Invited to be part of the NXTP Fellowship network. Receiving a monthly stipend, mentorship, and access to the firm's resources.

AGI House Fellow

06/2025

Selected to spend the summer at the AGI House in California, working on my projects.

Líder Estudar - Estudar Foundation - NGO

09/2021 - 05/2026

The most competitive scholarship in Brazil. 0.04% acceptance rate. Paid for my master's program.

Teaching Initiation Award - UFPB

11/2019

Awarded for the work done in the tutoring of Boolean Algebra and Logic Circuits class.